PRIME MINISTER'S OFFICE
REGIONAL ADMINNIRATION AND LOCAL GOVERNMENT
FORM TWO REGIONAL MOCK EXAMINATION
ARUSHA REGION
AGRICULTURE
MARKING SCHEME

SECTION A: 15 MARKS

01.

Questions	i	ii	iii	iv	v	vi	vii	viii	ix	x
Answers	D	A	C	C	B	D	B	D	C	A

--- 10 MARKS

02.

LHS A	i	ii	iii	iv	v
LHS B	C	D	A	E	B

--- 05 MARKS

SECTION B: 40 MARKS

Q3 (a) Critical stages when crops demand significant amount of water

(i) Germination / establishment stage
(ii) Flowering stage
(iii) Grain, seed, tuber formation --- 04 MARKS
(iv) Fruit formation stage.

(b) How cover crops and mulching retain soil moisture

(i) Reduce evaporation by covering the soil surface
(ii) Improve water infiltration into the soil
(iii) Reduce surface run-off
(iv) Lower soil temperature
(v) Conserving moisture --- 06 MARKS
(vi) Prevents direct exposure of the soil to the sunlight

04 (a) Four types of suckers as a propagation material in banana.

(i) Maiden suckers
(ii) Sword suckers
(iii) Water suckers
(iv) Piper suckers

- - - - - 04 MARKS

(b) Five maturity indicators for banana fruits ready to be harvested.

(i) Fruit size :- Mature bananas are bigger and well filled compared to immature ones.

(ii) Fruit shape :- Immature bananas have angular or sharp edges, matured have round shape with smooth edges.

(iii) Peel colour :- It changes from deep green to light green or slightly yellow as fruit mature

(iv) Pupl colour :- It changes from cream to yellow or orange yellow when matured

(v) Age of the bunch :- Banana bunches are usually ready for harvest 9-18 months after planting depending on the variety

- - - - (06 MARKS)

05

Data

Spacing = 70cm X 30cm

Area = 1 hectare ---------- 01 MARKS

Recall

Plant population = $\frac{\text{Area of the farm}}{\text{Spacing}}$ ---------- 01 MARKS

Converstion for spacing

1m = 100cm

? = 70cm

= 0.7m

1m = 100cm

? = 30cm ---------- 01 MARKS

= 0.3m

Converstion of Area

1ha = $10,000m^2$

Thus,

Plant Population = $\frac{10,000m^2}{0.7m \times 0.3m}$

= $\frac{10,000m^2}{0.21m^2}$ ---------- 01 MARKS

= 47,619 plants/ha.

---------- 01 MARKS

∴ The plant population = 47,619 plants/ha

Total = 05 MARKS

05 (b) Three banana diseases

(i) Banana bacterial wilt

(ii) Bunchy Top disease / Banana Bunchy Top Virus (BBTV)

(iii) Black Sigatoka

(iv) Cigar-end Rot

(v) Anthracnose — — — any three = 03 MARKS

(c) Banana insects

(i) Banana weevil

(ii) Nematodes

(iii) Vermins — — — any two = 02 MARKS

06. Five importance of animal housing

(i) It protects the animal from bad weather

(ii) It provides protection from theft and attack from dangerous wild animals.

(iii) It provides clean and comfortable environment

(iv) It reduces possibilities of diseases and mortalities

(v) It allows for efficient management of livestock

(vi) Housing promotes good welfare

— — — 05 MARKS.

6 (b) Five characteristics of good animal house.

(i) Adequate space
(ii) Adequate light during day hours
(iii) Good ventilation
(iv) Proper drainage
(v) Cleanliness and hygiene
(vi) Freedom to exhibit natural behaviour
(vii) Freedom to movement, stand, stretch and lie down.
(viii) Visual and social contact with other animals
(ix) Security from predators

Any five = 05 MARKS

07 (a) Factors to consider when selecting site for vegetable farming

(i) Fertile soil with good drainage
(ii) Suitable climate
(iii) Availability of water for irrigation.
(iv) Accessibility and infrastructure
(v) Low incidence of pest and diseases
(vi) Security of the area.

05 MARKS

(b)

CROP PLANT	SCIENTIFIC NAME	FAMILY NAME
Chinese Cabbage	Brassica rapa	Brassicaceae
Tomato plant	Lycopersicon esculentum	Solanaceae
onion plant	Allium cepa	Amaryllidaceae
Cassava plant	Manihot esculenta	Euphorbiaceae
Irish potato	Solanum tuberosum	Solanaceae

08 (a) Four equipments that should be necessary in poultry house.

(i) Feeders
(ii) Drinkers
(iii) Perches
(iv) Broody coops
(v) Carrying crest
(vi) Laying nest

any 04 = 04 MARKS

(b) (i) Free range system

Advantages

- Low investment
- Birds get varieties of foods as insects, grasses and grains
- Natural feeding and behaviour
- Less labour required
- Reduced occurance of vices
- Birds get alot of exercise

— — — — — 02 MARKS

(ii) Semi-intensive system

Advantages

- Less food wastage and space
- Low disease Spread
- Easier egg collection and management
- Minimal vices as egg-eating and pecking
- Low mortality rate
- Easy identification of sick birds

— — — — — 02 MARKS

(iii) Intensive System

Advantages

- Easy collection of manure
- Easy collection of eggs
- Easy cleanliness
- Easy identification and management of birds.

09. (a) Economic importance of horticulture
- Source of food in human being
- Source of income
- Source of foreign exchange
- Create employment opportunities
- Have aesthetic value to the environment
- Source of raw materials for various industries

---- 05 MARKS

(b) Important resources needed by farmers to start vegetable production

(i) Land
(ii) Labour
(iii) Capital
(iv) Water availability and quality
(v) Managerial skills.

---- 05 MARKS

SECTION C

10. Knowledge of various fields in contribution to Agricultural development in Tanzania

INTRODUCTION

Any relevant introduction - - - - - 03 MARKS

MAIN BODY

(i) Biology :- Understanding causes, symptoms and control methods of pests and diseases for crops and animals.

(ii) Chemistry :- Essential for soil test and soil analysis, manufacturing and application of fertilizer and other agrochemicals.

(iii) Physics :- Essential for manufacturing and operating various farm machine and implements as tractors, planters and ~~harvestors~~. harvestors.

(iv) Computer science :- Essential for precision farming in controlling and monitoring farm operations using computer integration

(v) Mathematics :- Operations as addition, substraction, division and multiplication help to calculate plant population and other required measurement as spacing,

(vi) Geography :- Understanding Climate and weather patterns helps farms to plan and make decisions on types of crops and breeds of animals to raise, but also to have timely planting to minimize risks.

(vii) Business studies :- Essential to determine loss and profit in farm entreprise, and entrepreneurial skills on how to operate entreprises

(viii) Food and Human nutrition :- Essential for food preservation and storage to ensure stability of food security

(ix) Historia ya Tanzania na maadili :- Encourages good work ethics, willingness to follow rules and guidelines, cultural value and sustainable farming

(x) Language studies :- Languages as English and Swahili are used as media of communication

— — — — @ 01 MARKS = 10

CONCLUSION

Any relevant conclusion — — — — — 02 MARKS